Methods Practice SAC2

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Notes + CAS	10m Reading | 1h 40m Writing	58 marks

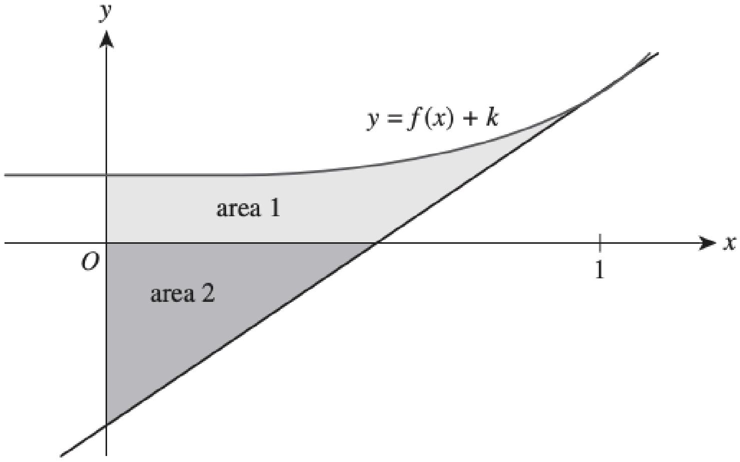
Question 1 (10 marks)
Neap 2021 Exam 2 Q1

Consider the function f with the rule $f(x) = x^3$.

a. Find the equation of the tangent at the point where $x = 1$. 1 mark

b. Find the area bound by $y = f(x)$, the tangent line at $x = 1$ and the x-axis. 3 marks

The graph of $y = f(x) + k$, where $k \in R^+$, and the tangent line at the point where $x = 1$ are sketched below.



c. Find the coordinates of the x-intercept of the tangent line in terms of k. 2 marks

d. Area 1 is the area bound by the curve $y = f(x) + k$, the tangent line at $x = 1$ and the coordinate axes. Area 2 is the area bound by the tangent line at $x = 1$ and the coordinate axes.

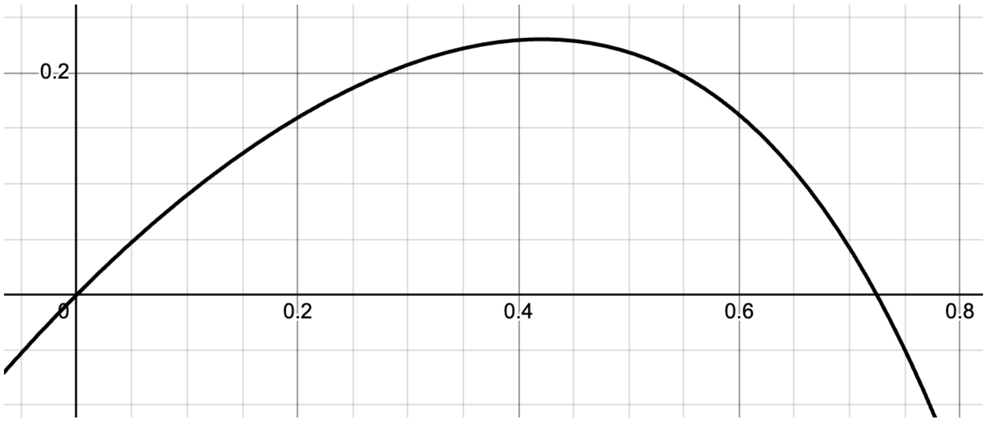
Find the value of k so that area 1 = area 2. 4 marks

Question 2 (12 marks)
Unlike quadratics, cubics and quartics, which all have formulas for their roots (x-ints), quintic polynomials (polynomials of degree 5) like:

$$f(x) = -x^5 - x^2 + x$$

have no general formula – they may have roots (x-ints) that can only ever be approximated.

A portion of the graph of f is shown below:



Consider the obvious root $x = 0$, and the unknown root to its right $x = p$.

a. Using Newton's method, choose a suitable initial guess x_0 to find the value of p. Fill out the table correct to 4 decimal places. (2 marks)

x_0	
x_1	
x_2	
x_3	

b. With reference to your iterations list (x_0, x_1, …), or otherwise, justify how you can be certain that some estimate x_n is giving p rounded to 4 decimal places. (2 marks)

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c. Hence, give the value of p correct to 4 decimal places.

If you are not confident that your x_3 correctly gives p to 4 d.p, do more iterations of Newton's method. (1 mark)

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Furthermore, if we only have an approximation for p, the area given by the integral:

$$A = \int_0^p f(x)\,dx$$

can also only be approximated.

d. Using 5 trapeziums and your previous 4 d.p approximation of p from (b), approximate the area A correct to 4 decimal places. (2 marks)

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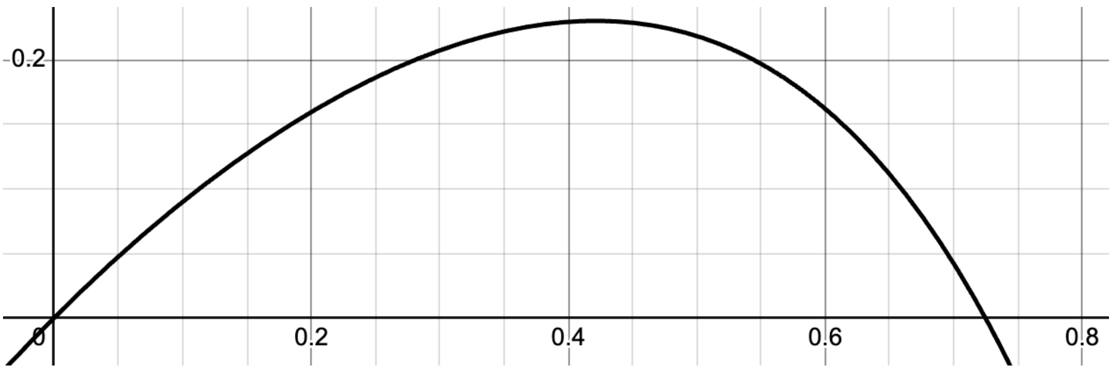
e. On the graph of f below, provide a rough sketch what your 5 area-approximating trapeziums look like. (2 marks)



f. Using your result from (d) and your previous 4 d.p approximation of p from (b), approximate the average value of f over the interval $x \in [0,\ p]$, correct to 4 decimal places. (2 marks)

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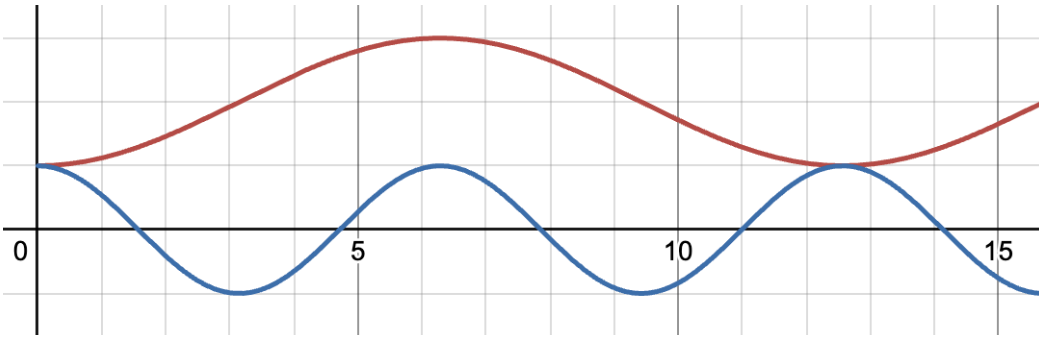
Question 3 (13 marks)

Consider the family of functions given by f and g, where:

$$f: [0, \infty) \to R,\ f(x) = -cos(ax) + 2 \qquad \text{and} \qquad g: [0, \infty) \to R,\ g(x) = cos(x)$$

a. Given $a = \frac{1}{2}$, the graphs of f and g appear as:



i. With $a = \frac{1}{2}$, state the general solution for $f(x) = 1$, using $n_1 \in Z$. (1 mark)

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ii. State the general solution for $g(x) = 1$, using $n_2 \in Z$. (1 mark)

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iii. By comparing the general solutions of both f and g, or otherwise, find the x-values of the points of intersection, x_1 and x_2, of f and g where $x_1, x_2 \in [0, 15]$. (2 marks)

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iv. Hence, or otherwise, find the area bounded by the graphs of f and g between $x = x_1$ and $x = x_2$. (2 marks)

b. Now consider $a = \frac{5}{3}$.

i. State the general solution for $f(x) = 1$, using $n_1 \in Z$. (1 mark)

ii. By comparing the general solutions of both f and g, or otherwise, state the x-values of the points of intersection of f and g, where $x \in [0, \infty)$ (3 marks)

When $a \in R$, the graphs of f and g will only ever have infinitely many intersections, or a single intersection at $x = 0$.

Consider when a is rational, $a \in Q$, such that $a = \frac{p}{q}$ where $p, q \in Z$.

c. Prove that when $a \in Q$ there will always be infinitely many intersections of f and g. (2 marks)

d. Hence, or otherwise, give a value of a such that there is only one intersection of f and g. (1 mark)

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Question 4 (15 marks)
Consider the function f given by:

$$f\colon [0,\ 4] \rightarrow R,\ f(x) = x\,cos(x)$$

a.

i. Find the derivative, $\frac{d}{dx}(\,x\,sin(x)\,)$. (1 mark)

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ii. Hence, or otherwise, show that an antiderivative of the the function f can be given as: (3 marks)

$$\int x\,cos(x)\,dx = cos(x) + x\,sin(x).$$

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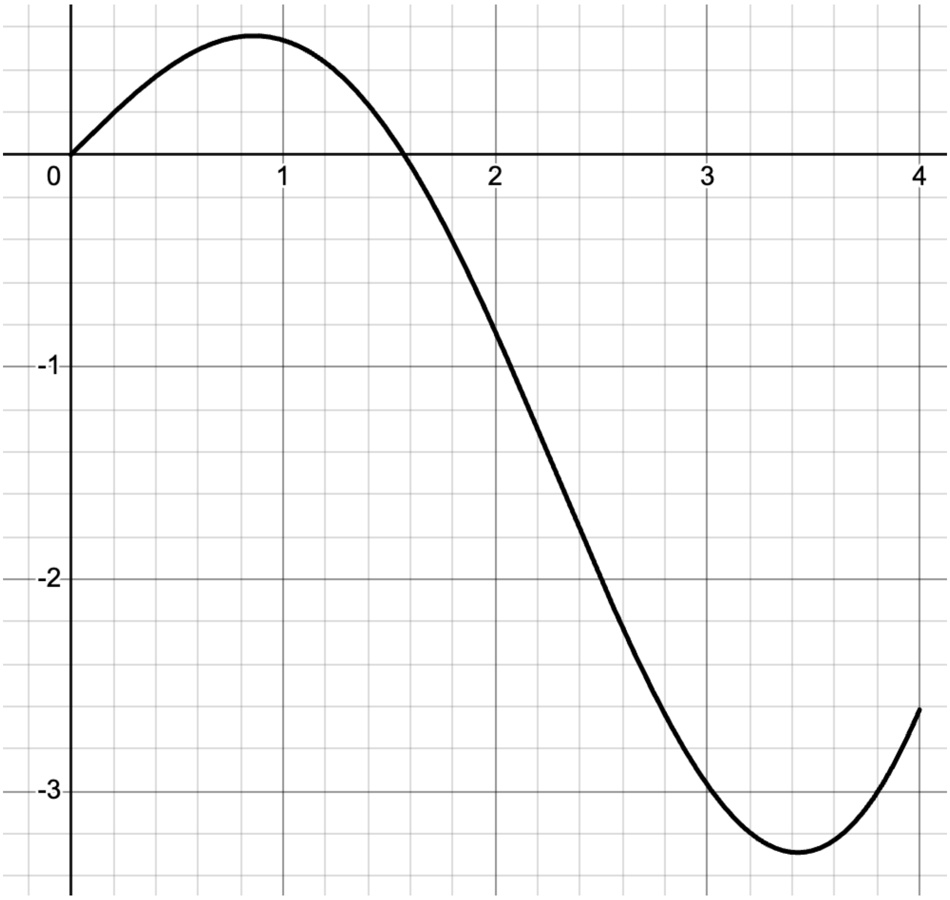
b. On the graph of f below, excluding those at $x = 0$, sketch a tangent **and** a normal line that pass through the origin, $(0, 0)$. (2 marks)



c.

i. Find the equation of the aforementioned <u>tangent</u> line. (2 marks)

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ii. Find the area bounded by this tangent line and the curve f. (2 marks)

d.

i. Find the equation of the aforementioned <u>normal</u> line, correct to 3 decimal places. (2 marks)

ii. Find the area bounded by this normal line and the curve f, correct to 3 decimal places. (3 marks)

Question 5 (10 marks)

Consider the function f:

$$f(x) = a - \frac{a}{5(x-2)}$$

where $a \in R$.

a. State the rule and domain of the inverse function, f^{-1}. (2 marks)

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b. Find the value of a such that f and f^{-1} intersect infinitely many times. (1 mark)

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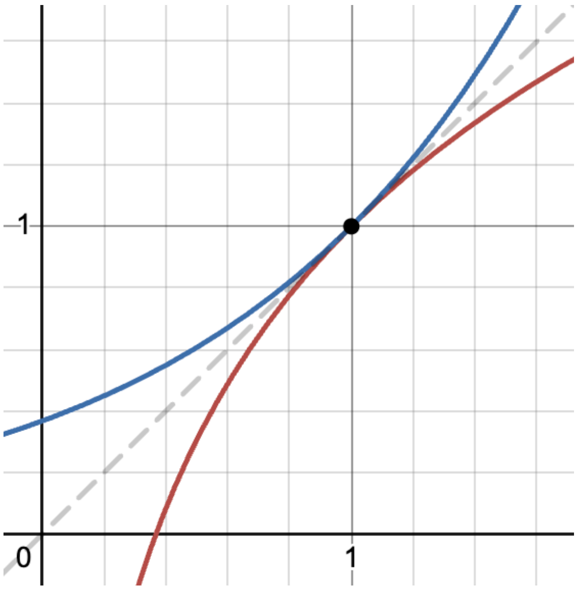
It can be shown that in order for f and f^{-1} to 'just touch' one another, like:



it needs to be at some x-value such that:

$$\begin{cases} f(x) = x \\ f'(x) = 1 \end{cases}$$

Note: we should solve this system of equations simultaneously for x and a.

c. Find the value(s) of a such that f and f^{-1} intersect only once, correct to 3 decimal places. (2 marks)

d. Using your result from (c) and observing the graphs of f and f^{-1}, or otherwise, state the values of a such that f and f^{-1} intersect exactly twice, correct to 3 decimal places. (3 marks)

Consider the case where $a \in (0,\ 1)$.

e. By observing the graphs of f and f^{-1}, or otherwise, state the area bounded by f and f^{-1} as $a \to 0^{+}$

That is, what value does this area approach as a approaches 0? (1 mark)